PHASE-3 PROJECT REPORT | DATA ANALYTICS

PHASE-3 REPORT
Data Analytics Track | Tasks 11–15

Name	Vaishnavi Raut
Track	Data Analytics – PulseIQ
GitHub Repository	https://github.com/vaishnaviraut629-de/pulseiq
Git Tag	phase-3
Commit	54a081a
Status	Phase 3 Complete

1. Task Completion Summary

Task	Status	Outcome / Evidence
Task 11: P&L; and expenses	Completed	Created DuckDB view v_pnl using detail accounts only. April 2025 revenue: 71,075,300; expense: 64,761,237; difference = 0. Excluded group accounts 4000 and 5000 to prevent double counting. Evidence: Task11_PnL_71L.png
Task 12: Invoices and receivables	Completed	Created four aging buckets (0–30, 31–60, 61–90, 90+ days). Total buckets matched open invoices. Evidence: Task12_Aging.png
Task 13: Cross-department impact	Completed	Aggregated GL by month and Budget by department-month before joining. This avoided double counting. Evidence: Task13_Grain.png
Task 14: AI signals dashboard	Completed	Built AI signals page; labelled outputs as AI-generated / Predicted and sorted Money at Risk descending. Evidence: Task14_AI.png
Task 15: Executive scorecard	Completed	Created an executive-friendly one-page scorecard with 8 KPIs. Evidence: Task15_Scorecard.png

2. Validation and Proof of Functionality

DuckDB validation: April 2025 revenue was 71,075,300 and expense was 64,761,237; difference = 0. May 2025 revenue of 60,215,000 was also checked. Detail-account totals matched, with group accounts excluded. Receivables aging buckets added up to open invoices.

Git validation: git status showed “nothing to commit, working tree clean.” git push showed “Everything up-to-date.” Commit 54a081a is pushed to the correct repository.

3. Mini Challenge: Why Did Cost Rise in August?

Page / Check	Finding
P&L; page	August expense was 50.5L versus July 52.8L; vendor share was higher in August.
Vendor page	NimbusCloud was among the top three vendors, with a 12.3L Cloud Hosting bill.
Cross-department view	Engineering department D01 showed increased overtime in August.
Scorecard	August company-health score was 72; the Cost indicator was red.

Conclusion: The August cost pattern was associated with the NimbusCloud hosting bill and increased Engineering overtime. I investigated it using four dashboard pages in under five minutes.

4. Self-Check: Can I Join Sources at the Right Grain and Prove Totals?

Answer: Yes. I aggregated GL data to month level and Budget to department-month level, joined on month and department, and reconciled the result against detail accounts to avoid double counting.

In Phase 2, I used find_bad.py to remove four bad payroll rows, including PI-X0001 (9,999,999). The cleaned DuckDB total matched the answer key (difference = 0). Engineering payroll for January 2024 was manually checked: 50,000 + 60,000 + 55,000 = 165,000.

5. Key Learnings

• Data cleaning and correct grain are essential; group accounts can cause double counting.

• Aggregating before joining helps prevent inflated totals.

• Dashboard numbers should be reconciled with underlying DuckDB data.

• Predictions should be clearly labelled as Predicted or AI-generated.

• A clean Git status and pushed code help make the project reproducible.

6. Problems Encountered and Resolutions

Issue	Resolution
SQL command error in CMD	CMD does not execute SQL directly. I ran the query through Python and DuckDB instead.
“Nothing to commit”	The tasks were already committed in 54a081a. Git push confirmed everything was up to date.
Repository name confusion	git remote -v confirmed vaishnaviraut629-de/pulseiq as the correct repository.
P&L; double counting	Excluded group accounts 4000 and 5000, then validated the detail-account total (difference = 0).

7. Next Phase Plan

Phase 4 will focus on Packaging, Release and Self-service (Tasks 16–20). I will prepare a clear README and requirements.txt, make dashboards easier for non-technical users, push the phase-4 tag, record a demo video, and keep predicted risk outputs clearly labelled.

8. Screenshots and Supporting Evidence

Task	Evidence filename
Task 11 – P&L;	Task11_PnL_71L.png
Task 12 – Receivables	Task12_Aging.png
Task 13 – Grain	Task13_Grain.png
Task 14 – AI signals	Task14_AI.png
Task 15 – Scorecard	Task15_Scorecard.png

Repository and Submission Details

GitHub: https://github.com/vaishnaviraut629-de/pulseiq
Git tag: phase-3 | Commit: 54a081a | Status: Phase 3 Complete
Submission note: phase-3 tag pushed; report prepared as one PDF for programme submission.

End of Phase-3 Report